Experiment 10

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AIM: The aim is to create data visualizations, such as line charts and doughnut charts, for a student performance analysis system using JavaScript and Chart.js.

PROCEDURE:

Step 1: Set Up Your HTML File Create an HTML file to define the structure, including canvas elements where the charts will be rendered, and link the Chart.js library.

HTML (index2.html**):**

HTML

```
<!DOCTYPE html>

<html>

<head>

   <title>Student Performance Charts</title>

   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

   <style>        body { font-family: Arial, sans-serif; background-color: #f4f4f4; text-align:

center; }        h2 { color: #222; }        canvas {            display: block;            margin: 20px

auto;            background-color: #fff;            border: 1px solid #ddd;            border-radius:

6px;            box-shadow: 0 2px 6px rgba(0, 0, 0, 0.1);

       }

   </style>

</head>

<body>

   <h2>Student Performance Analysis</h2>


   <h3>Marks Progress (Line Chart)</h3>

   <canvas id="lineChart" width="400" height="200"></canvas>
```

```
   <h3>Subject-wise Performance (Doughnut Chart)</h3>

   <canvas id="doughnutChart" width="400" height="200"></canvas>


   <script src="script1.js"></script>

</body>

</html>
```

Step 2: Create the JavaScript File for Charts Create a JavaScript file to define the datasets for test marks and subject distribution, and initialize the charts using the Chart.js constructor.

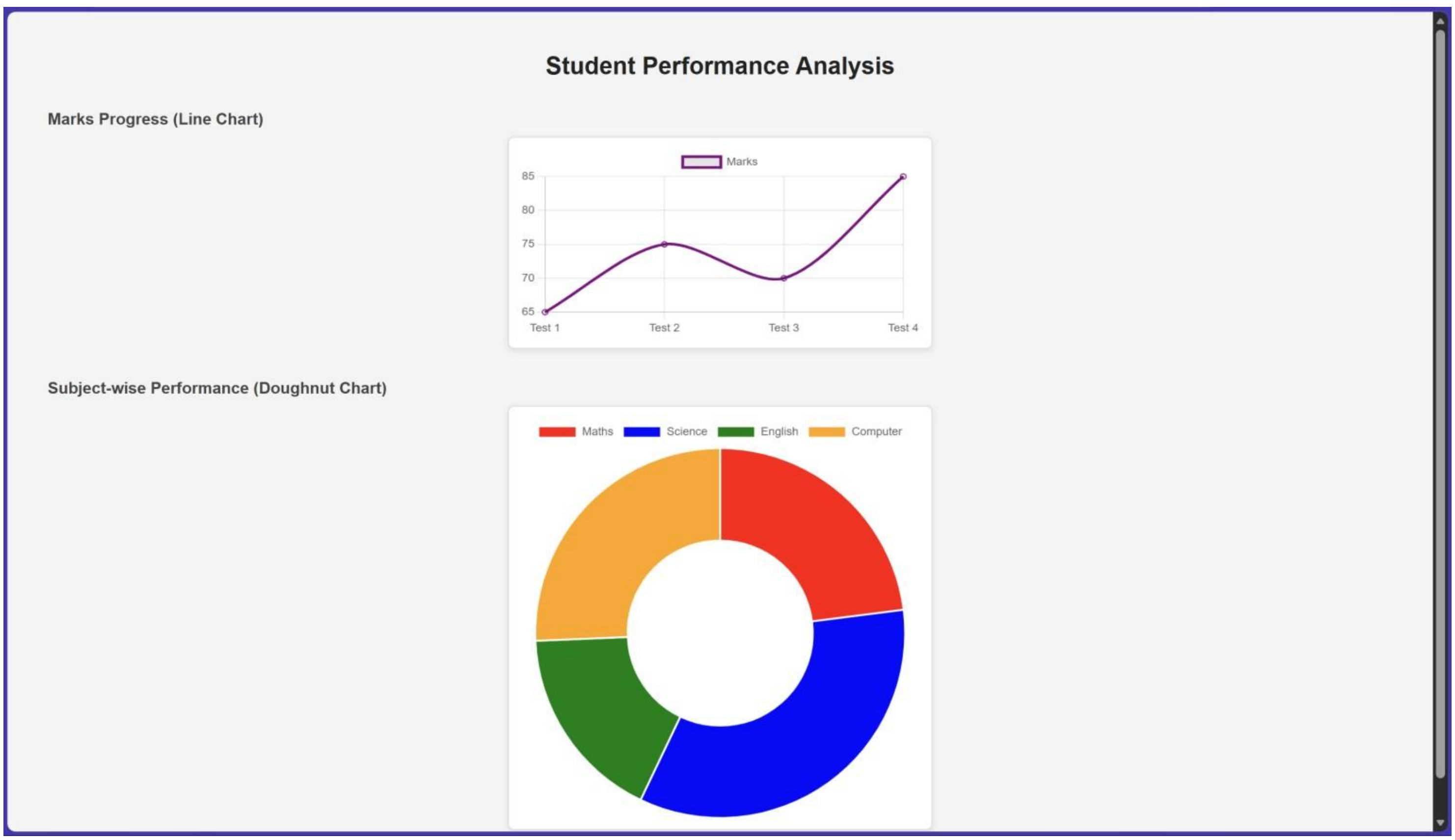
JavaScript (script1.js**):**

JavaScript

```
// Line Chart - Marks Progress const lineCtx =

document.getElementById('lineChart').getContext('2d'); const lineChart =

new Chart(lineCtx, {     type: 'line',     data: {        labels: ['Test 1', 'Test 2',

'Test 3', 'Test 4'],        datasets: [{           label: 'Marks',           data: [65, 75,

70, 85],           borderColor: 'blue',

          fill: false,

          tension: 0.3

       }]

   },

   options: { responsive: true }

});


// Doughnut Chart - Subject Performance const doughnutCtx =

document.getElementById('doughnutChart').getContext('2d'); const doughnutChart =

new Chart(doughnutCtx, {     type: 'doughnut',     data: {        labels: ['Maths', 'Science',

'English', 'Computer'],        datasets: [{           label: 'Marks Distribution',           data:

[85, 75, 80, 90],           backgroundColor: ['red', 'blue', 'green', 'orange']

       }]
```

```
  },
  options: { responsive: true }
});
```

OUTPUT:



RESULT: The data visualizations for student performance were successfully created using JavaScript and Chart.js, providing a clear graphical representation of marks progress and subject-wise distribution.